

HARISH NAVNIT

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Summary

- 7+ years software engineering experience with customer-facing roles working on integrations at Dialpad and a research, solutions and engineering hybrid role at Imperial Global Singapore.
- 4+ years owning technical solution delivery across customer-facing and early-stage engineering roles.
 - Simultaneously acting as the architect, PM, and engineer taking a research initiative through prototype, live stakeholder demos, and signed MoUs at Imperial Global Singapore.
 - Scoped requirements directly with customers and AEs, coordinated cross-functional delivery across platform, QA, and product teams, shipped enterprise integrations that strengthened the PLG arm of the Dialpad product.
 - Earlier in career, was hired as the first engineer at Anapi. Owned all releases, hiring, and technical direction with no dedicated PM layer. Directly met with customers to deliver bespoke sandboxes and solutions.
- Experienced in driving technical adoption and clearing blockers across complex, multi-stakeholder engagements.
 - Built strategic enterprise integrations that delivered \$700k additional revenue/Q at Dialpad.
 - Co-ordinated cross-functional delivery across platform, QA, product and professional service to save a \$200-300k/year at-risk customer by designing a technical workaround and shipping outside normal sprint cycles.
 - Established sign-off processes and communication protocols that improved cross-team delivery.
 - Actively managing multi-party research deployments at Imperial Global Singapore with both local and overseas partners.
- Hands-on familiarity with enterprise AI tooling ecosystems including frontier LLM providers (OpenAI, Anthropic, Google Gemini) and adjacent knowledge management and workflow automation platforms.
- Architected end-to-end agentic AI solutions integrating third-party data sources, enterprise workflows, and custom AI agents to drive technical adoption and manage proof-of-concept work.
 - Built an agentic AI penetration testing platform, scaling from early research prototype to signed MoUs and early pilot deployments.
 - Invited to showcase the platform at the Singapore British High Commission during SICW 2025.
 - Converted interested partners into executive engagements that resulted in strategic and commercial agreements.
- Deep cloud expertise across AWS, Azure & GCP.
 - Built real-time event pipelines, serverless functions, and IaC policies.
 - Authored a mocking library for Azure Python SDK, deployed internally at ColorTokens.
 - Deployed & scaled Dialpad's integrations platform to several thousand DAU's on the Google Cloud Platform (GCP).
- Strong focus on developer productivity, containerization and infrastructure.
 - Hands-on with Docker, Kubernetes and microservice architecture scaling.
 - * Authored build-scripts and maintained cross-platform build systems.
 - * Experienced in setting up legacy CI/CD pipelines and transitioning to modern cloud based runners such as Github Actions.
- Seasoned Pythonista with strong JS/TS fluency.
 - Contributed upstream improvements to ElevenLabs MCP server tools.
 - Contributed a Semgrep context provider for Sourcegraph's OpenCtx.
 - Authored Python SDK's for Dialpad & Drone CI, in addition to the mocking library for Azure's Python SDK.
 - Participated in Google Summer of Code, contributing to prominent KDE projects.

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EXPERIENCE

Imperial Global Singapore | *Software Research Engineer, AI Solutions*

Apr. 2025 – Present

Imperial College London's interdisciplinary research centre in NUS, Singapore.

- Managing end-to-end delivery of **AI-first, Cybersecurity** research prototypes across multi-stakeholder collaborations by scoping requirements, co-ordinating with programme directors and liaising with external partners.
- Conducted customer discovery sessions and executive briefings to identify AI use cases, translating complex agentic workflows into actionable business outcomes for non-technical stakeholders.
- Partnered with RRIS Singapore to architect a telephonic Montreal Cognitive Assessment (MoCA) tool using ElevenLabs conversational agents for clinical deployment across public hospitals in Singapore.
- Leading the engineering and product development for an end-to-end automated pen-testing platform for medical devices and hospital networks using agentic coding tools and MCP servers.
 - * Identified critical security gaps and HIPAA compliance violations in real-world medical devices & servers, viz. patient monitors, DICOM servers.
 - * Showcased the platform at British High Commission in Singapore during Singapore International Cyber Week and secured MoUs with interested partners.

Dialpad | *Software Engineer, Integrations*

Aug. 2021 – Sept. 2023

- Developed strategic microservice integrations in **Python & Vue.js**, deployed on the Google Cloud Platform(**GCP**).
- Delivered \$700k additional revenue/Q across Salesforce, Hubspot, Zendesk & Servicenow integrations.
- Served as a technical point of contact for enterprise customers, scoping bespoke solutions and translating requirements into engineering deliverables to drive adoption and reduce churn.
- Led technical recovery of a \$200-300k/year at-risk enterprise account by designing a tailored workaround, coordinating cross-functional delivery, and presenting the solution directly to the customer.

ColorTokens | *Member of technical staff*

Jan. 2020 – Aug. 2021

- Implemented the architectural re-design of XShield, an enterprise **micro-segmentation** platform, by migrating from a monolith to a **micro-service** based design.
- Prototyped, built & maintained **Python/Flask** micro-services for discovery and real-time monitoring of customer assets on public clouds, viz **AWS** and **Azure**, forming a robust *Cloud Security Posture Management(CSPM)* offering. Further developed,
 - * an interval based polling mechanism for asset discovery.
 - * a real-time event ingestion pipeline for platform events using serverless(**AWS Lambda**) functions and custom webhooks.
 - * Infrastructure-as-Code(**IaC**) polices using Cloud Custodian to facilitate policy enforcement & compliance.

Anapi | *Senior Software Engineer*

Aug. 2018 – Sept. 2019

- As the first engineering hire, owned all product releases, technical architecture decisions and vendor co-ordination, shaping crucial early product roadmap.
- Additionally, responsible for hiring, onboarding and mentoring junior engineers on the team.
- Designed a robust digital insurance **API** using **Node.js** deployed via **Docker** builds to **AWS**, to help online businesses enable on-demand protection through appropriate insurance products.
- Architected bulk policy purchase and claims ingestion pipelines, ensuring **data sanitization** for regulatory and auditory compliance.

Spectral Insights (acq. **Evident Scientific**) | *Software Engineer*

Mar. 2017 – Jun. 2018

- Integrated software components of a medical whole slide scanning device, viz. a **C++** image processing pipeline, **PyQt** GUI for hardware interface and **Django** cloud service.
- Maintained a **Python** cloud interface to publish the derived analytics for a pathology slide scan.

INTERNSHIPS

Google Summer of Code | tiny.cc/gsoc-16

Jun. 2016 - Sept. 2016

- Reworked KDE Plasma's legacy public transport applet written in **C++/Qt** to provide a convergent UI across devices of varying form factors.

Google Summer of Code | tiny.cc/gsoc-15

Jun. 2015 - Sept. 2015

- Ported Kopete, a multi-protocol instant messaging client, from **C++/Qt** (Qt4) to the newer **QtQuick 2.0** based KDE frameworks 5 API.

PROJECTS

Spikes | spikes.dev

NextJS, React, Python, Typescript, LiteLLM

- AI observability platform that measures token spend against engineering impact.
- Provides a unified gateway to monitor usage and route requests to fallback low-cost models.

OpenCtx | openctx.org

LLM, Typescript, VSCode extension

- Implemented an *LLM* context provider for [Sourcegraph Cody](#), which allows it to view [Semgrep](#) findings for a project and respond to user queries appropriately.
- Demo: youtu.be/Cm_9QGrbMdA

Mazure | github.com/tinvaan/mazure

Azure, Mocks, Python, Testing

- A stateful and dynamic *Python* library that enables easy mocking of Azure API calls in unit tests.
- Supports *function decorators* and *context manager* modes of usage.
- Demo: youtu.be/WCLCNlima0M

ASR Metrics | github.com/ImperialGlobalSingapore/asrmetrics

NLTK, OpenAI Whisper, Typescript, Python

- Automatic Speech Recognition (ASR) dashboard, with speaker diarization support, interactive transcripts editor and detailed evaluation metrics.

Drone CI SDK | github.com/tinvaan/drone-python

Automation, CI/CD, Python

- Python client for the [Drone CI](#) public API.

Postag | github.com/tinvaan/postag

AIS, AWS, NMEA, Python

- Generates a deployable AWS lambda function for Satellite AIS message tagging.

CONFERENCES AND TECHNICAL PRESENTATIONS

conf.kde.in | tiny.cc/kde-cmake

IIT Guwahati, 2020

Hosted a workshop on *"Introduction to CMake"*

FOSSASIA summit | youtu.be/Df9hFzZl7Tc

Singapore, 2018

Delivered a presentation on *"Convergent applications using the Kirigami UI framework"* introducing a modern C++/Qt framework for responsive and cross-device UI designs.

KDE developer sprints

Switzerland, 2015 - 2017

- Invited to participate in the Randa Meetings | randa-meetings.ch
- Invited to the inaugural WikiToLearn & KDE developer's sprint held at the IdeaSquare lab at CERN

EDUCATION

Singapore University of Technology and Design

2023 - 2024 | 4.2/5.0

Master of Science in Technology and Design (Cybersecurity)

Top of batch

Amrita School of Engineering

2012 - 2016 | 3.6/5.0

Bachelor of Technology in Computer Science and Engineering

SKILLS

Languages: C, C++, C++/Qt, QML, Python, HTML, CSS, JavaScript, TypeScript, Bash, Shell Scripting

Cloud & Infrastructure: Google Cloud Platform (GCP), AWS (EC2, S3, RDS, Lambda), Azure, Docker, Kubernetes, Vagrant, CI/CD (Github Actions)

Frameworks: FastAPI, Flask, Django, React, Vue, Node.js, Express.js, QtQuick, Bootstrap, Tailwind, Scikit-learn, Pydantic

Databases: MySQL, PostgreSQL, Sqlite, MongoDB, AWS RDS, Google Cloud Datastore

Tools: SVN, Git, GitHub, Gitlab, Bitbucket, Unix Shell, VS Code, Datadog, Sentry, Wireshark, Tableau, Weka

AI Tools: GPT, Github Copilot, GPT-Codex, Cursor, Codex, Claude, Claude Code, Claude Code SDK, Anthropic AI, ElevenLabs, OpenAI Whisper, Gemini, Gemini Studio, Gemini SDK, Langchain, Pydantic AI, Instructor